

Question 1 - McCall model: comparative statics

Please see the provided code.

1. Under the baseline parameterization, the reservation wage is 49.17.

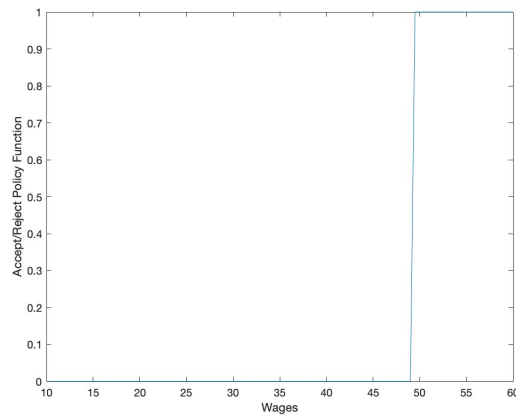


Figure 1: Policy Function

2. See Table 1.

	Reservation Wage
b=10	47.66
b=30	50.88
b=50	55.45

Table 1: Different unemployment benefits

3. See Table 2.
4. With log utility (and thus risk-averse preferences) rather than linear utility (and thus risk-neutral preferences), the reservation wage decreases to 46.74.

	Reservation Wage
$\beta = 0.8$	38.59
$\beta = 0.99$	54.33

Table 2: Different discount factors benefits

Question 2 - McCall model with infrequent offers

1. In this environment, wages continue to be a state variable but now, whether the unemployed worker receives a wage offer is a state as well.

The value function for not receiving a wage offer is:

$$U = b + \beta [\mathbb{E}[\phi V(w') + (1 - \phi)U]]$$

The value function for receiving a wage offer is:

$$V(w) = \max_{\text{Accept, Reject}} \left\{ \frac{w}{1 - \beta}, U \right\}$$

Note that there is no firing or job destruction, so once the worker accepts a wage offer, they'll receive that wage forever.

2. Please see the provided code. For $\phi = 0.7$, the reservation wage decreases to 47.42 (or 47.5 if you are using value function iteration on the discrete grid), which is lower than the baseline specification where $\phi = 1$. Note that we can still directly compute the reservation wage as shown in Discussion 9 by finding the root (γ) to the following equation:

$$\gamma = b + \beta \left(\phi \beta \sum_{i=1}^N \max \left\{ \frac{w_i}{1 - \beta}, \gamma \right\} p_i + (1 - \phi) \gamma \right)$$